

Encapsulation Is Not Encryption

What Flannel does, and does not, protect in a K3s cluster

VXLAN

Encapsulated, readable

WireGuard

Encapsulated + encrypted

HTTPS

Encrypted by the app

01

Flannel Backends & Encryption

What the default K3s setup does, and does not, protect

THE ONE FACT TO REMEMBER

Default K3s Pod traffic is not encrypted

```
k3s server --flannel-backend=vxlan      # the default
```

	Encapsulation	Encryption
VXLAN	Yes	No
WireGuard	Yes	Yes
IPsec	Yes	Yes

VXLAN provides encapsulation, not confidentiality.

A tunnel, but not a private one



What VXLAN adds

- Outer node IPs so the underlay can route it
- A UDP + VXLAN header (VNI identifies the network)
- About 50 bytes of overhead

What VXLAN does NOT add

- No encryption of the inner packet
- No authentication of the sender
- No integrity check against tampering

What anyone on the node network can read

Outer IP 192.168.1.10 → 192.168.1.20

UDP 8472 + VXLAN header

Inner IP 10.42.1.10 → 10.42.2.10

TCP 80

GET /login HTTP/1.1 user=alice&pass=...

All of this is plain text

- Source and destination Pod IPs
- Ports and protocols
- HTTP headers, cookies, tokens
- Request and response bodies

Unless the application itself uses TLS, VXLAN exposes the payload to the underlay.

Which backends encrypt?

Backend	Encapsulation	Encryption	Node port	Notes
vxlan	Yes (VXLAN)	No	UDP 8472	Default. Best for seeing packets in labs
host-gw	No (plain routes)	No	n/a	Nodes must share an L2 network
wireguard-native	Yes (WireGuard)	Yes	UDP 51820	Kernel WireGuard; interface flannel-wg
none	n/a	n/a	n/a	Bring your own CNI (Calico, Cilium)

Removed from K3s

The older ipsec and wireguard (userspace) backends. Use wireguard-native instead.

Need IPsec?

Use a CNI that offers it, e.g. Cilium, with `--flannel-backend=none`.

Choosing encrypted or unencrypted

Unencrypted (e.g. to inspect traffic)

```
# /etc/rancher/k3s/config.yaml
flannel-backend: vxlan

# or on the command line
k3s server --flannel-backend=vxlan
```

Encrypted Pod-to-Pod traffic

```
# /etc/rancher/k3s/config.yaml
flannel-backend: wireguard-native

# or on the command line
k3s server \
  --flannel-backend=wireguard-native
```

Watch out

- Set the backend on every server node and choose it before building the cluster
- Changing it later means restarting all nodes and disrupts Pod traffic while tunnels rebuild
- Do not use wireguard-native if your goal is to observe unencrypted Pod traffic

02

Two Kinds of "Encryption"

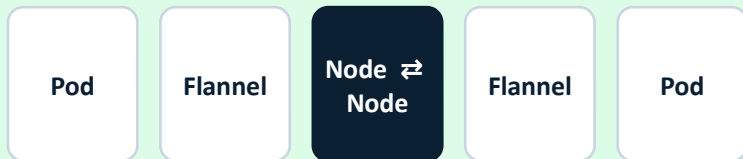
Network-level vs application-level, and why they are independent

Who is doing the encrypting?

1

Network encryption

Controlled by the Flannel backend



- Protects ALL Pod traffic between nodes
- Transparent: apps need no changes
- Only node-to-node; same-node traffic is not encrypted

2

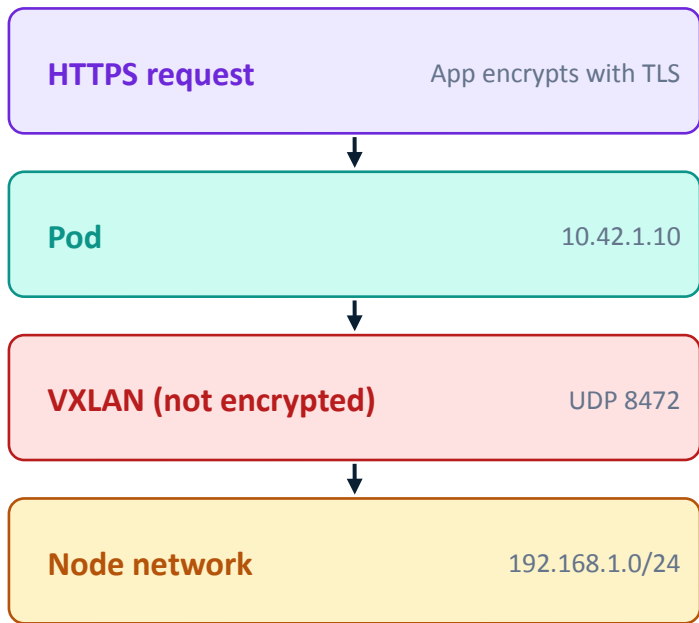
Application encryption

Controlled by the app or a service mesh

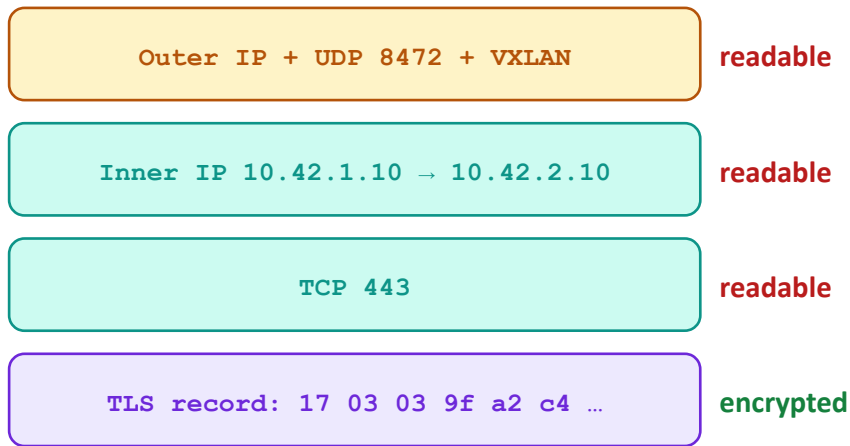


- End to end, Pod to Pod
- Works with any CNI or backend
- Needs certificates in each app, or mTLS via a mesh

Unencrypted tunnel, encrypted payload



On the wire an observer sees



The payload is protected by HTTPS; metadata (IPs, ports) is still visible.

What can the node network see?

Flannel backend →

vxlan

wireguard-native

App uses
plain HTTP

Everything readable

IPs, ports, headers, bodies

Encrypted between nodes

Readable only on the node itself

App uses
HTTPS / mTLS

Payload encrypted

Pod IPs and ports visible

Encrypted twice

Defence in depth

K3s control-plane traffic is always TLS

Control plane

kubelet ↔ API server, etcd peers, kubect

Always TLS (K3s-managed certs)

Pod network

Pod ↔ Pod across nodes

Depends on Flannel backend

Application

HTTP, gRPC, database protocols

Depends on the app or mesh

Choosing vxlan does not weaken cluster management traffic; it only affects Pod data.

See plain-text VXLAN with tcpdump

```
# 1. Two nginx Pods, ideally on different nodes
kubectl create deployment web --image=nginx \
  --replicas=2
kubectl get pods -o wide

# 2. On the node hosting the target Pod
sudo tcpdump -ni eth0 -A udp port 8472

# 3. From another node's Pod, send a request
kubectl exec -it <web-pod-on-node1> -- \
  curl -s http://<web-pod-ip-on-node2>
```

Expected in the capture

IP 192.168.1.10 > 192.168.1.20:
VXLAN

IP 10.42.1.10 > 10.42.2.10: TCP

GET / HTTP/1.1
Host: 10.42.2.10

<title>Welcome to nginx!</title>

Both the request and the response are readable.

Switch to wireguard-native and compare

```
# On each server: set backend and restart
sudo vi /etc/rancher/k3s/config.yaml
#   flannel-backend: wireguard-native
sudo systemctl restart k3s
# agents: sudo systemctl restart k3s-agent

# Verify the interface
ip -d link show flannel-wg

# Capture again
sudo tcpdump -ni eth0 -A udp port 51820
```

Compare the captures

vxlan · UDP 8472

Inner Pod IPs visible
HTTP request readable
nginx HTML readable

wireguard-native · UDP 51820

Only node IPs visible
Payload is random bytes
No Pod IPs, no HTTP

Match the backend to the goal

Teaching / lab: show packets

`vxlan`

Easiest to capture and explain

Trusted private network, apps use TLS

`vxlan`

Simple, low overhead, payload already protected

Untrusted links: cloud, edge, WAN

`wireguard-native`

Encrypts all cross-node Pod traffic transparently

Compliance: encrypt everything

`wireguard-native + mTLS`

Network and application layers together

KEY TAKEAWAYS

A tunnel is not a vault

vxlan

is the K3s default: encapsulated, not encrypted.

wireguard-native

encrypts Pod traffic between nodes.

App TLS

is independent of Flannel and protects the payload end to end.

Labs

use vxlan so students can read the packets with tcpdump.